# Supplementary Materials for the paper “Two distinct causal beliefs organizing intuitions about mind, action, and body” by Joseph Outa and Shari Liu

For questions, please email the corresponding author Joseph Outa jouta1@jh.edu.

**Table of contents**



All data and analysis scripts required to reproduce the results in this paper, including helpful .html reports of the analyses, are available at:
https://github.com/liulaboratory-experiments/mind-action-body

## 1: Experiment 1 - Representational Similarity Analysis

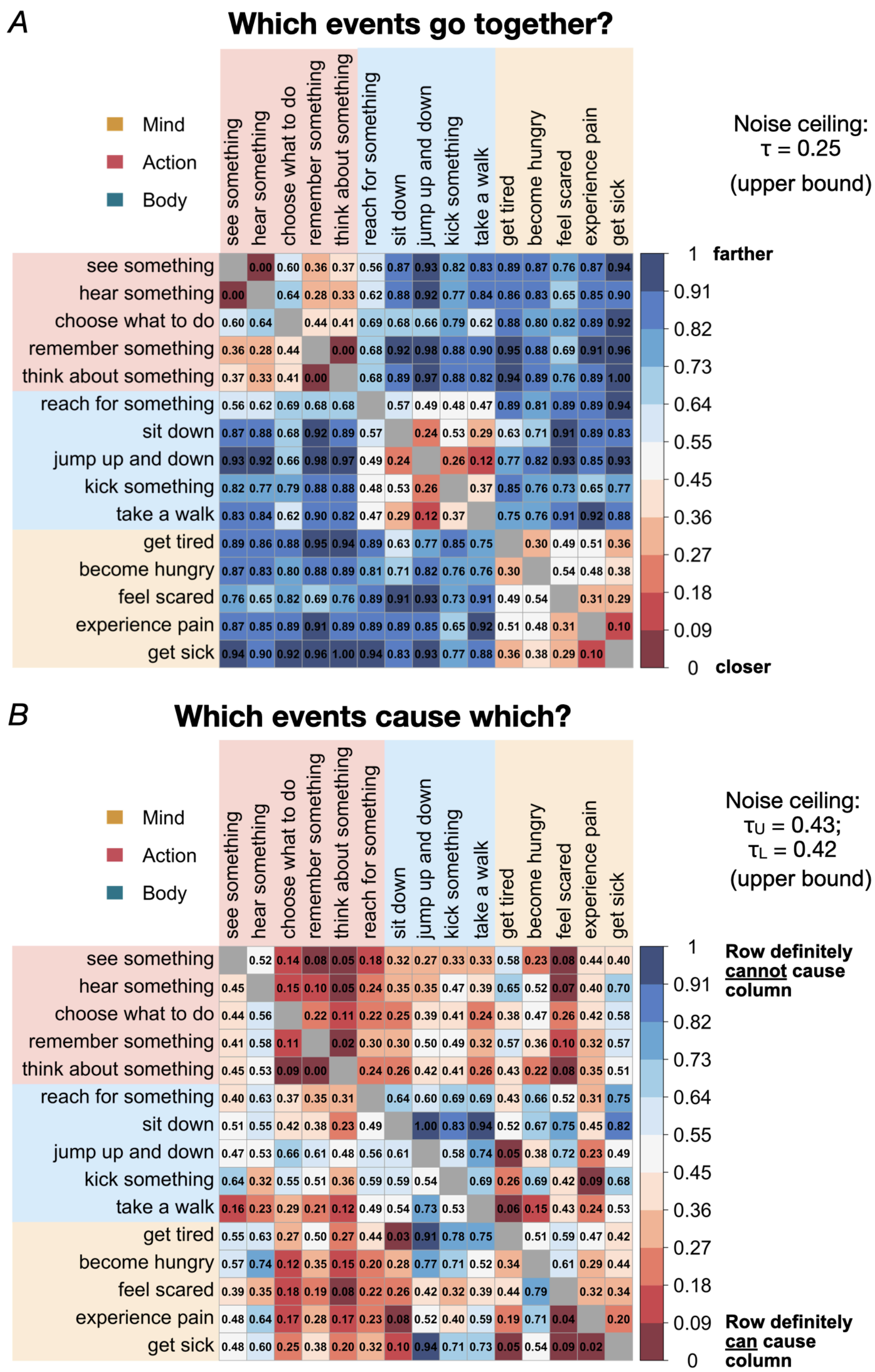


**Figure S1:** Results for Experiment 1 with numeric values included.

### *1.1 Generating Freesort Task RDM*

The output of the Freesort Task was a position vector with x and y coordinates for the final position of each of the 15 items on the canvas. We calculated the euclidean distance between each pair of items, using the formula $\sqrt{(x_{itemA} - x_{itemB})^2 + (y_{itemA} - y_{ItemB})^2}$ . We then min-max normalized the resulting distance by subtracting the minimum euclidean distance and dividing this by the range, as shown in this formula: $\frac{(euclidean\ distance - min(euclidean\ distance)}{max(euclidean\ distance) - min(euclidean\ distance)}$. We called this final value the *freesort distance*. For each participant these values were organized into a 15 by 15 matrix—a representational dissimilarity matrix (RDM)—with each value in the matrix corresponding to the freesort distance between pairs of items. We called this the *Freesort RDM.* A group-averaged RDM was created by  averaging the RDMss across subjects; this is depicted in Figure 1B of the main manuscript.

### *1.2 Generating Causal Pairs Task RDM*

The output of the Causal Pairs Task was an integer ranging from 0 (definitely not) to 100 (definitely yes) (corresponding on the final location of the slider knob). We inverted these response values, which mapped ratings of 100 to 0 and ratings of 0 to 100 using the formula $1 - response / 100$. This operation converted the measure into a dissimilarity measure (with values close to 0 meaning causally close, and values close to 1 meaning causally distant). Finally, we min-max normalized the values (as was done in the Freesort Task), and called the final value the *causal pairs distance*. These were also organized into a 15 by 15 matrix to form the subject-specific *causal RDM*, which was averaged across participants to form the mean causal RDM, depicted in Figure 1B of the paper.

### *1.3 Comparing Observed RDMs to Model RDMs*

Given these RDMs (one per participant per task) we compared them to different explanatory models, each expressed as an RDM with distinctive representational geometries for organizing the 15 items (see Figure S1, x-axis). The 4 models were as follows:

(1) *Psychological and physical*: Organizes the items into 2 categories: mental items (5 items) and physical items (10 items, actions and bodily events). This is in line with the predictions of Intuitive Dualism, which argues that we intuitively draw a clear boundary between events of the physical world and events of the mental world (Berent et al., 2022).
(2) *Mind, action, body model*: Organizes the items into 3 categories with 5 items per category: mental items, and actions, and bodily items. This is similar to (1), except that we further distinguish between physical overt behaviors (actions), and physiological processes internal to the body.
(3) *Fine-grained mind, action, and body*: We built this model to make one further distinction within each domain proposed in (2). This RDM organizes the items into 6 categories: perceptual events (2 items, seeing and hearing), cognitive events (3 items, remembering, thinking, and choosing), object-directed actions (2 items, reaching and kicking), non-object-directed actions (3 items, walking, sitting and jumping), phasic changes that are more drawn out in time (3 items, sick, tired, hungry), versus ones which begin and end more abruptly (2 items, scared, pain).

(4) *Phrase embeddings*: This model is based on embeddings from the Universal Sentence Encoder model, which captures the similarity in semantic meaning between all 15 items in terms of similar shared features, and graded differences between the items (Cer et al., 2018). This is based on the hypothesis that phrases which appear in similar contexts have similar meanings, and can thus be captured by vector based representations derived from large scale corpora using machine learning techniques (McClelland & Rogers, 2003). To generate model predictions for this model, we generated phrase embeddings for each of the 15 items using the USE model. Each embedding was a 512-dimensional vector capturing the semantic content of the phrase. To get a similarity measure, we computed the cosine similarity between each pair of phrase embeddings by taking the inner product of the two vectors and dividing by the product of their magnitudes: $cos\theta = \frac{A \cdot B}{|A||B|}$. This produced a value ranging from -1 (completely dissimilar) to 1 (completely similar). Since our analysis required a dissimilarity measure, we transformed the cosine similarity by rescaling it using the formula $\frac{1- cosine\ similarity}{2}$. This made the values range from 0 (very similar) to 1 (very dissimilar).

For each of these four models, we computed Kendall's $\tau$ between each participant's RDM and the theoretical RDM expressed by the model. For RDMs from the Freesort Task, we took the values from one half of the off-diagonal values, since the responses in this task were necessarily symmetrical; for RDMs from the Causal Pairs Task, we took the values from both halves of the off-diagonal matrix, since people gave ratings about both whether one item could cause a second, and vice versa. We plotted the participant-level correspondences with the model RDMs alongside their respective noise ceilings (**Figure S1**). Then, we compared the distribution of Kendall's $\tau$ across participants to 0 (no relationship between the model and the data), to measure how well a given model accounts for responses across participants in each task. We also ran a linear mixed effects model using the lmer() function from the lmerTest package in R, model specification τ ~ model_RDM + (1 | subject id), to compare the fit of a baseline model—the mind, action and physiology model—to the remaining 3 model RDMs (the fine-grained model, the physical and psychological and the USE phrase embeddings model. The full results of these analyses are shown in **Tables S1-3**.

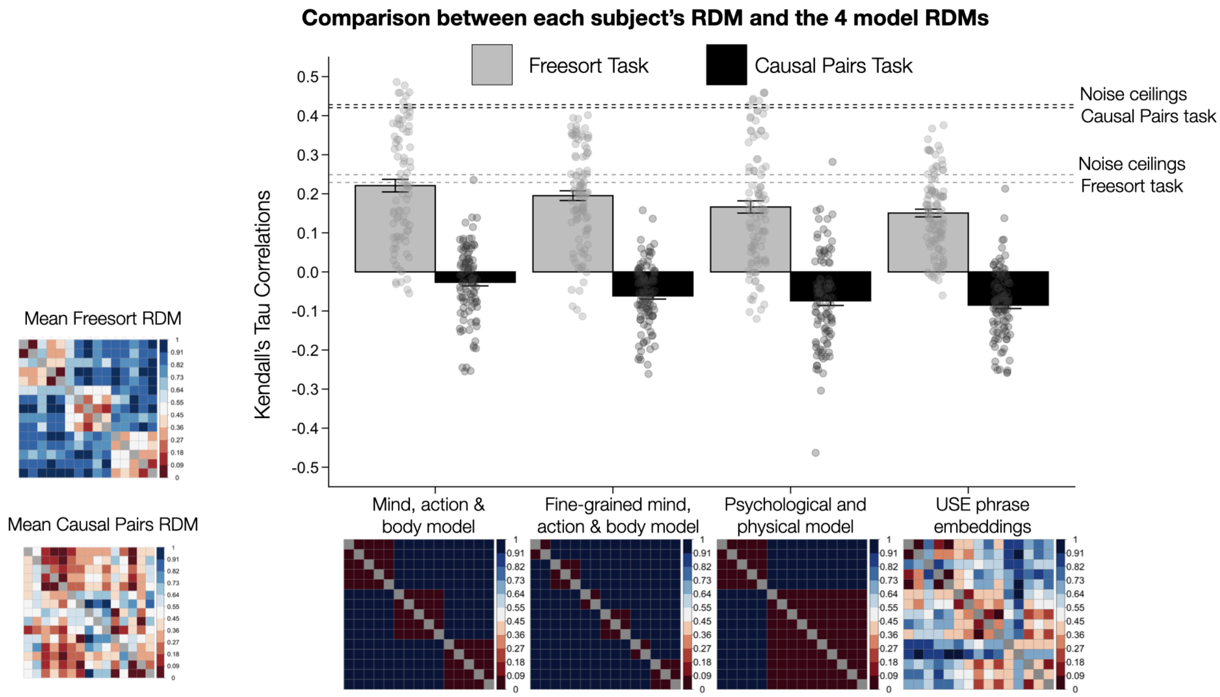


**Figure S2:** Kendall's $\tau$ correlations between the four models and average RDMs in the Freesort and Causal Pairs Task of Experiment 1. Most of the variance in the Freesort Task RDM was captured by the Mind, action and body model (this correlation reached the task's noise ceiling), followed by the Fine-grained mind, action and body model, the Psychological and physical model, and finally the Universal Sentence Encoder (USE) embedding model. Responses from the Causal Pairs Task were negatively correlated with all four models, Mind, action and body model performing closest to 0. The mean Freesort and Causal Pairs RDMs from subjects' responses are provided for reference in the bottom left. Individual points indicate subject-level correlations, error bars indicate 95% confidence intervals. *** p < .001, two-tailed.

**Table S1. Linear mixed-effects model comparing Kendall's correlations between the FreesortTtask and each of the four models.**

Model 1: tau ~ model_RDM + (1 | subject_id)

| Parameter | Coefficient | SE | t | df_error | p | Effects | Group |
|---|---|---|---|---|---|---|---|
| Mind action and Body (baseline) | 0.27 | 0.01 | 18.50 | 398 | 0 | fixed | |
| Fine grained mind action physiology model (vs baseline) | -0.08 | 0.01 | -5.66 | 398 | 0 | fixed | |
| Psychological and physical model (vs baseline) | -0.10 | 0.01 | -7.58 | 398 | 0 | fixed | |
| USE phrase embeddings model (vs baseline) | -0.12 | 0.01 | -9.01 | 398 | 0 | fixed | |
| SD (baseline) | 0.11 | | | | | random | subject_id |
| SD (Observations) | 0.09 | | | | | random | Residual |

**Table S2. Linear mixed-effects model comparing Kendall's correlations for the Mind, action and physiology model across the Freesort vs Causal Pairs asks.**

Model 2: tau ~ task + (1 | subject_id)

| Parameter | Coefficient | SE | t | df_error | p | Effects | Group |
|---|---|---|---|---|---|---|---|
| Freesort Task (baseline) | 0.71 | 0.07 | 10.09 | 198 | 0 | fixed | |
| Causal Pairs Task | -1.42 | 0.10 | -14.27 | 198 | 0 | fixed | |
| SD (baseline) | 0.00 | | | | | random | subject_id |
| SD (Observations) | 0.71 | | | | | random | Residual |

**Table S3. Linear mixed-effects model comparing Kendall's correlations between the four models for the Causal Task.**

Model 3: tau ~ model_RDM + (1 | subject_id)

| Parameter | Coefficient | SE | t | df_error | p | Effects | Group |
|---|---|---|---|---|---|---|---|
| Mind, action & body model (baseline) | -0.03 | 0.01 | -2.67 | 398 | 0.01 | fixed | |
| Fine grained mind, action and body model | -0.03 | 0.01 | -3.49 | 398 | 0.00 | fixed | |
| Psychological and physical model | -0.05 | 0.01 | -5.26 | 398 | 0.00 | fixed | |
| USE phrase embeddings model | -0.06 | 0.01 | -5.85 | 398 | 0.00 | fixed | |
| SD (baseline) | 0.07 | | | | | random | subject_id |
| SD (Observations) | 0.07 | | | | | random | Residual |

## 2: Experiment 1 - Order effects

All participants did both the Freesort and Causal Pairs Tasks in a randomized order. To examine whether task order influenced the structure of participants' ratings, we examined the correlation between responses of participants who completed the tasks first versus second, and compared it to their within-order correlations. For each task, we took two groups (participants who did the task first, those who did it second), and divided each group into odd and even subject IDs. Then we computed the average representational dissimilarity matrices (RDMs) for each subgroup (i.e. first group, odd; first group, even; second group, odd; second group, even), resulting in 4 mean RDMs, 1 per subgroup. We then computed a within-order similarity metric for each of the two groups by calculating their Kendall's $\tau$ correlation (for the Freesort Task, first odd and first even $\tau = 0.59$, and second odd and second even $\tau = 0.58$). We then compared these within-order similarities to cross-order similarities obtained by correlating RDMs across the two groups, using the same odd-even splits (first odd and second even $\tau = 0.615$;first even and second odd $\tau = 0.509$). We then averaged these values to produce a mean within-order and a mean across-order similarity for each task. For the Freesort Task, the mean within-order correlation ($\tau = 0.58$) was comparable to the mean cross-order correlation ($\tau = 0.56$). A similar result was obtained for the Causal Pairs Task (mean within-order $\tau = 0.76$, mean cross-order $\tau = 0.76$). This suggests that task order did not substantially influence participants' behavior in either task.

## 3: Experiment 2 through 6 - Item-selection Procedure

We used the quantitative distances generated from Experiment 1 (N = 101), combined with a pilot sample (N = 50) to select items used in Experiments 2 - 6.

### *3.1 Experiment 2 (Explanation), 3 (Counterfactuals) and 5 (Magical v Ordinary Intervention)*

For each target item, we designed a choice set consisting of a target item, a causal origins (CO) item, and a causal chains (CC) item (**Table S4**). The causal origins (CO) item was the item most relevant to the target in terms of the causal origin representation, and the causal chain (CC) item was most relevant to the target in terms of the causal chain representation. We defined relevance to the target in terms of distance to the target in freesort space relative to causal pairs space based on the average RDM from each task. We call this the *joint distance*. Since we hypothesized that causal origin representation is measured by the freesort task and the causal chains representation is measured by the causal pairs task, we reasoned that a choice item is relevant to the causal origin representation if the item is closer to the target in freesort space and further to the target in causal pairs space. Similarly, a choice item is relevant to the causal chain representation if it is closer to the target in causal pairs space and further in freesort space.

As such, the CO item was the item with the smallest joint distance (i.e. is closest in freesort space and simultaneously farthest in causal pairs space) to the target item. For example, for the target item “jump up and down”, a candidate “sit down” had a freesort distance of 0.35 and a causal pairs distance of 0.84, leading to a *joint distance* (i.e. the freesort minus causal pairs distance) of −0.49. This was the lowest joint distance among all possible pairings of “jump up and down” and other items; for example, another candidate item, “think about something” had a joint distance of 0.34. Thus, “sit down” became the CO item. We imposed a constraint that the CO item be in the same domain as the target item; this was already true for 13 of 15 of the target items; the 2 exceptions were “choose what to do” (i.e. an item in the “mind” domain) whose lowest joint distance item was “jump up and down” (i.e. an item in the “action” domain); and “get sick” (i.e. an item in the “physiological” domain) whose lowest joint distance item was “sit down” (i.e. an item in the “action” domain).

For choosing the CC items for Experiments 2, 3 and 4, we looked at potential preceding items to the target (i.e. those that were rated as highly likely to cause the target). The CC item was the item with the largest joint distance to the target, that is, the item closest in causal pairs space to the target item and farthest in freesort space to the target item. We imposed a constraint that the most CC item be in a different domain from the target item, though this was already true for 14 of the 15 items (i.e. all items except for “choose what to do”, whose largest joint distance items were a tie between “hear something” and “feel scared”).

| Target and Choice Items | | | |
|---|---|---|---|
| Domain | Target Item | CC item (CP distance, Freesort distance, joint distance) | CO item (CP distance, Freesort distance, joint distance) |
| **mind** | **see something** | take a walk (0.2, 0.58, 0.38) | hear something (0.42, 0.18, -0.24) |
| | **hear something** | take a walk (0.25, 0.58, 0.33) | see something (0.44, 0.18, -0.27) |
| | **choose what to do** | experience pain (0.2, 0.57, 0.36) | remember something (0.15, 0.36, 0.22) |
| | **remember something** | take a walk (0.24, 0.61, 0.37) | think about something (0.07, 0.15, 0.08) |
| | **think about something** | get sick (0.25, 0.65, 0.4) | remember something (0.08, 0.15, 0.08) |
| **action** | **reach for something** | become hungry (0.21, 0.56, 0.35) | kick something (0.54, 0.4, -0.15) |
| | **sit down** | experience pain (0.13, 0.58, 0.45) | jump up and down (0.54, 0.3, -0.23) |
| | **jump up and down** | see something (0.29, 0.6, 0.32) | sit down (0.83, 0.3, -0.52) |
| | **kick something** | think about something (0.37, 0.6, 0.23) | sit down (0.7, 0.42, -0.29) |
| | **take a walk** | think about something (0.26, 0.58, 0.32) | sit down (0.77, 0.33, -0.44) |
| **body** | **get tired** | jump up and down (0.1, 0.54, 0.43) | feel scared (0.42, 0.4, -0.02) |
| | **become hungry** | take a walk (0.18, 0.55, 0.37) | feel scared (0.67, 0.44, -0.22) |
| | **feel scared** | see something (0.12, 0.54, 0.42) | get tired (0.51, 0.4, -0.11) |
| | **experience pain** | kick something (0.13, 0.51, 0.38) | get tired (0.43, 0.41, -0.02) |
| | **get sick** | see something (0.38, 0.62, 0.25) | get tired (0.39, 0.34, -0.06) |

**Table S4**: Items used in Experiment 2, 3, and 4. The first two columns list the target events and their domains. The remaining two columns list the two options (the “CC item” vs “CO item”) associated with that target, and their freesort and causal pairs distance from the target. These distances were computed from combined group-averaged responses of Experiment 1 (N = 101) and a pilot (N = 50).

### *3.2 Experiment 4 (Inference v Prediction)*

The CO items for Experiment 4 were selected in the same way as the items for Experiments 2, 3 and 5 (see above). Because we ask people to make predictions about future events in this study (rather than asking them to go backwards, in Experiments 2 3 and 5), to select CC items, we looked for items that were likely effects of the target items, i.e. with the biggest joint distance relative to the target (**Table S5**).

| Target and Choice Items | | | |
|---|---|---|---|
| Domain | Target Item | CC item (CP distance, Freesort distance, joint distance) | CO item (CP distance, Freesort distance, joint distance) |
| mind | see something | feel scared (0.12, 0.54, 0.42) | hear something (0.44, 0.18, -0.27) |
| | hear something | feel scared (0.12, 0.48, 0.36) | see something (0.42, 0.18, -0.24) |
| | choose what to do | feel scared (0.28, 0.54, 0.26) | hear something (0.5, 0.49, -0.01) |
| | remember something | feel scared (0.15, 0.52, 0.37) | hear something (0.51, 0.32, -0.19) |
| | think about something | feel scared (0.14, 0.54, 0.4) | hear something (0.49, 0.34, -0.14) |
| action | reach for something | experience pain (0.29, 0.58, 0.29) | take a walk (0.62, 0.41, -0.21) |
| | sit down | think about something (0.26, 0.61, 0.35) | jump up and down (0.83, 0.3, -0.52) |
| | jump up and down | get tired (0.1, 0.54, 0.43) | take a walk (0.64, 0.24, -0.41) |
| | kick something | experience pain (0.13, 0.51, 0.38) | take a walk (0.59, 0.33, -0.26) |
| | take a walk | get tired (0.12, 0.53, 0.42) | jump up and down (0.63, 0.24, -0.39) |
| body | get tired | sit down (0.09, 0.46, 0.37) | become hungry (0.44, 0.3, -0.14) |
| | become hungry | think about something (0.19, 0.58, 0.39) | feel scared (0.55, 0.44, -0.11) |
| | feel scared | think about something (0.14, 0.54, 0.4) | become hungry (0.67, 0.44, -0.22) |
| | experience pain | sit down (0.13, 0.58, 0.45) | become hungry (0.61, 0.4, -0.21) |
| | get sick | sit down (0.14, 0.55, 0.41) | become hungry (0.49, 0.36, -0.12) |

**Table S5**: Items used in Experiment 4. The first two columns list the target events and their domains. The remaining two columns list the two options (the “CC item” vs “CO item”) associated with that target, their freesort distances, causal pairs distances and joint distances to the target. These distances were computed from group-averaged responses from Experiment 1 (N = 101) and a pilot (N = 50).

### *3.3 Experiment 6 (Biological Reasoning vs Causal Triads)*

For Experiment 6, we first used the Causal Pairs RDM to first find potential causal chains (A → B → C), for which A and C are from the same domain, but not strongly causally related (**Table S6**). We came up with 9 chains that had this property. Item B, the item with the largest average joint distance to A and C, became the CC item. Then to choose a CO item, we identified the item with the smallest average joint distance to the A and B.

| | Target items | | Middle item in forced choice response | | | |
|---|---|---|---|---|---|---|
| domain | First target | Second target | CC item | Mean joint distance | CO item | Mean joint distance |
| mind | see something | hear something | take a walk | 0.581 | remember something | 0.052 |
| | choose what to do | hear something | take a walk | 0.492 | see something | −0.126 |
| | remember something | hear something | feel scared | 0.458 | think about something | −0.086 |
| action | sit down | jump up and down | experience pain | 0.401 | take a walk | −0.568 |
| | jump up and down | sit down | get tired | 0.664 | take a walk | −0.398 |
| | take a walk | reach for something | see something | 0.545 | jump up and down | −0.330 |
| body | get tired | feel scared | choose what to do | 0.532 | become hungry | −0.118 |
| | become hungry | feel scared | think about something | 0.699 | get tired | −0.058 |
| | get sick | become hungry | remember something | 0.518 | experience pain | −0.052 |

**Table S6**: Items used in Experiment 6. The first two columns list the target events and their domains. The remaining four columns list the two options (the “CC item” vs “CO item”) associated with that target, and their joint distance from the target. These distances were computed from group-averaged responses from Experiment 1.

## 4: Demographic information and exclusion rates for each study

| | Experiment 1 (Measure) | Experiment 2 (Explanation) | Experiment 3 (Counterfactuals) | Experiment 4 (Inference v Prediction) | Experiment 5 (Magical v Ordinary Intervention) | Experiment 6 (Biological Reasoning v Causal Triads) |
|---|---|---|---|---|---|---|
| **N after exclusions** | 101 | 104 | 102 | 102 | 105 | 104 |
| **N excluded (reason)** | 1 (failed attention check) | 1 (failed attention check) | 3 (failed attention check) | 5 (failed attention check) | 0 | 0 |
| **Mean age (min, max)** | 38.06 (19, 73) | 40.10 (20, 80) | 41.07 (21, 70) | 42.73 (19, 79) | 41.5 (19, 79) | 42.61 (18, 75) |
| **Female** | 41 | 41 | 53 | 63 | 59 | 56 |
| **Male** | 60 | 63 | 52 | 38 | 45 | 45 |
| **Non-binary** | 1 | 3 | 1 | NA | NA | 1 |
| **White** | 73 | 74 | 82 | 79 | 85 | 80 |
| **Black/African American** | 17 | 9 | 8 | 8 | 6 | 8 |
| **Asian** | 8 | 15 | 12 | 13 | 12 | 10 |
| **Latino/Hispanic** | 2 | 1 | 1 | NA | NA | NA |
| **Other** | 7 | NA | 1 | 5 | 2 | 6 |

**Table S7.** Sample sizes, exclusions and demographic information (age, gender, ethnicity) across the 6 experiments.

## 5. Study demos

Demos for all studies are available below:
Experiment 1: https://liulaboratory-experiments.github.io/mbc/demo/experiment1
Experiment 2: https://liulaboratory-experiments.github.io/mbc/demo/experiment2
Experiment 3: https://liulaboratory-experiments.github.io/mbc/demo/experiment3
Experiment 4a: https://liulaboratory-experiments.github.io/mbc/demo/experiment4_inference
Experiment 4b: https://liulaboratory-experiments.github.io/mbc/demo/experiment4_prediction
Experiment 5a: https://liulaboratory-experiments.github.io/mbc/demo/experiment5_magical
Experiment 5b: https://liulaboratory-experiments.github.io/mbc/demo/experiment5_ordinary
Experiment 6a:
https://liulaboratory-experiments.github.io/mbc/demo/experiment6_biological_reasoning
Experiment 6b: https://liulaboratory-experiments.github.io/mbc/demo/experiment6_causal_triads